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Buffer choice and the clarity of a model protein: a teaching example

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Journal of Tutorial Examples · 2026

Abstract

We compare two buffers for a model protein and record how clear each sample looks after dilution. In this invented example, buffer A kept the protein solution clear for at least 30 minutes, while buffer B produced a slight haze. A longer incubation in buffer B increased the haze. The example shows how a short, well-recorded comparison can guide the next experiment.

Introduction

Choosing a buffer is often the first decision in a protein experiment. A buffer that keeps the protein soluble makes later measurements easier to interpret. This teaching example describes a simple comparison that a student can repeat with any model protein and two candidate buffers.

The aim is not to identify the best buffer for every protein. Instead, the example shows how to plan the comparison, record each observation with its conditions, and decide what to test next.

Methods

Sample preparation

The model protein was thawed on ice and diluted to 1 mg/mL in each buffer. Each dilution was mixed gently by pipetting and kept at 4 °C.

Observation

Each tube was held against a dark background and scored as clear, slightly hazy, or cloudy after 30 minutes and after 2 hours. Two people scored every tube independently.

Results

Buffer A kept the protein solution clear at both time points. Buffer B produced a slight haze after 30 minutes, and the haze increased after 2 hours.

Buffer	30 minutes	2 hours
Buffer A	Clear	Clear
Buffer B	Slightly hazy	Cloudy

Table 1. Invented observations for the teaching example.

Both observers gave the same score for every tube. No measurement of absorbance or particle size was made in this example.

Discussion

In this example, buffer A is the better starting point for later experiments. The increase in haze with time suggests that buffer B slowly destabilizes the model protein. A useful next experiment would repeat buffer B with a measured readout, such as absorbance at 340 nm, to confirm the visual scores.

Visual scoring is quick but subjective. Recording who scored each tube, and when, keeps the observation useful when it is compared with later measurements.

References

1. Example Author. A fictional guide to planning simple comparisons. Tutorial Press; 2025.
2. Demo Author. Recording observations so they can be reused. Journal of Tutorial Examples; 2024.